

STAT 325 - Design of Experiment Homework 1

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STAT 325 HW 1 1/15/26.

(1) $p(x) = \begin{cases} \frac{x}{10} & x=1, 2, 3, 4 \\ 0 & \text{otherwise} \end{cases}$

A. $p(Y=1) = 1/10, p(Y=2) = 2/10, p(Y=3) = 3/10, p(Y=4) = 4/10$

B. $E(X) = \sum x p(x) = \frac{1}{10} + 2(\frac{2}{10}) + 3(\frac{3}{10}) + 4(\frac{4}{10}) = \frac{30}{10} = 3$

C. $V(X) = E(X^2) - E(X)^2 = (\frac{1}{10} + 4(\frac{2}{10}) + 9(\frac{3}{10}) + 16(\frac{4}{10})) - 3^2 = 10 - 9 = 1$

D. $X_1, X_2 \sim \text{iid } p(x)$

$E(4X_1 - 3X_2) = 4E(X_1) - 3E(X_2) = 4(3) - 3(3) = 3$

$E \cdot \text{Var}(4X_1 - 3X_2) = 16(1) + 9(1) = 25$

In general we have $\text{Var}(aX_1 + bX_2) = E((aX_1 + bX_2) - (a\mu_1 + b\mu_2))^2$

$\text{Var}(aX_1) + \text{Var}(bX_2) + 2\text{Cov}(aX_1, bX_2)$

Since X_1, X_2 are ~~independent~~ iid. $\text{Cov}(aX_1, bX_2) = 0$ then

$= a^2 \text{Var}(X_1) + b^2 \text{Var}(X_2)$

A. $f(z) = \frac{0.72135}{z} = 0.72135$

(2) $f(y) = \begin{cases} \frac{0.72135}{y} & 1 \leq y \leq 4 \\ 0 & \text{otherwise} \end{cases}$ $P(1 \leq Y \leq 2) = \int_1^2 \frac{0.72135}{y} dy \approx 0.5$

B. $E(Y) = \int_1^4 y \left(\frac{0.72135}{y} \right) dy = 0.72135 (y) \Big|_1^4 = 0.72135 (4 - 1) = 2.16405$

C. $\text{Var}(Y) = \int_1^4 y^2 \left(\frac{0.72135}{y} \right) dy - (2.16405)^2 \approx 0.7210$

D. $E(2Y+1) = 2E(Y) + 1 = 2(2.16405) + 1 = 5.3281$

$E \cdot \text{Var}(2Y+1) = 4 \text{Var}(Y) = 4(0.7210) = 2.884$

Figure 1: Handwritten Work

Problem #1

Given PDF: $p(x) = \frac{x}{10}$ for $x = 1, 2, 3, 4$ (discrete distribution)

A. Probabilities

$$\begin{aligned}P(X = 1) &= \frac{1}{10} \\P(X = 2) &= \frac{2}{10} = \frac{1}{5} \\P(X = 3) &= \frac{3}{10} \\P(X = 4) &= \frac{4}{10} = \frac{2}{5}\end{aligned}$$

Verify: $\frac{1+2+3+4}{10} = \frac{10}{10} = 1$

B. Expected Value $E(X)$

$$\begin{aligned}E(X) &= \sum_{x=1}^4 x \cdot p(x) \\&= 1 \cdot \frac{1}{10} + 2 \cdot \frac{2}{10} + 3 \cdot \frac{3}{10} + 4 \cdot \frac{4}{10} \\&= \frac{1 + 4 + 9 + 16}{10} = \frac{30}{10} = 3\end{aligned}$$

C. Variance $V(X)$

First, find $E(X^2)$:

$$\begin{aligned}E(X^2) &= \sum_{x=1}^4 x^2 \cdot p(x) \\&= 1^2 \cdot \frac{1}{10} + 2^2 \cdot \frac{2}{10} + 3^2 \cdot \frac{3}{10} + 4^2 \cdot \frac{4}{10} \\&= \frac{1 + 8 + 27 + 64}{10} = \frac{100}{10} = 10\end{aligned}$$

$$\begin{aligned}V(X) &= E(X^2) - [E(X)]^2 \\&= 10 - 3^2 = 10 - 9 = 1\end{aligned}$$

D. $E(4X_1 - 3X_2)$

For independent random variables:

$$\begin{aligned}E(4X_1 - 3X_2) &= 4E(X_1) - 3E(X_2) \\&= 4(3) - 3(3) \\&= 12 - 9 = 3\end{aligned}$$

E. $V(4X_1 - 3X_2)$

For independent random variables:

$$\begin{aligned}V(4X_1 - 3X_2) &= 4^2V(X_1) + (-3)^2V(X_2) \\&= 16(1) + 9(1) = 25\end{aligned}$$

Problem #2

Given PDF: $f(y) = \frac{0.72135}{y}$ for $1 \leq y \leq 4$ (continuous distribution)

A. $f(2)$ and $P(1 \leq Y \leq 2)$

$$f(2) = \frac{0.72135}{2} = 0.360675$$

$$\begin{aligned} P(1 \leq Y \leq 2) &= \int_1^2 \frac{0.72135}{y} dy \\ &= 0.72135 \int_1^2 \frac{1}{y} dy \\ &= 0.72135 [\ln y]_1^2 \\ &= 0.72135 (\ln 2 - \ln 1) \\ &= 0.72135 \cdot \ln 2 \\ &= 0.72135 \cdot 0.6931 \approx 0.5000 \end{aligned}$$

B. Expected Value $E(Y)$

$$\begin{aligned} E(Y) &= \int_1^4 y \cdot \frac{0.72135}{y} dy \\ &= 0.72135 \int_1^4 1 dy \\ &= 0.72135 [y]_1^4 \\ &= 0.72135 (4 - 1) \\ &= 0.72135 \cdot 3 = 2.16405 \end{aligned}$$

C. Variance $V(Y)$

First, find $E(Y^2)$:

$$\begin{aligned} E(Y^2) &= \int_1^4 y^2 \cdot \frac{0.72135}{y} dy \\ &= 0.72135 \int_1^4 y dy \\ &= 0.72135 \left[\frac{y^2}{2} \right]_1^4 \\ &= 0.72135 \cdot \frac{16 - 1}{2} \\ &= 0.72135 \cdot \frac{15}{2} = 5.4101 \end{aligned}$$

$$\begin{aligned} V(Y) &= E(Y^2) - [E(Y)]^2 \\ &= 5.4101 - (2.16405)^2 \\ &= 5.4101 - 4.6831 = 0.7270 \end{aligned}$$

D. $E(2Y + 1)$

$$\begin{aligned} E(2Y + 1) &= 2E(Y) + 1 \\ &= 2(2.16405) + 1 \\ &= 4.3281 + 1 = 5.3281 \end{aligned}$$

E. $V(2Y + 1)$

$$\begin{aligned} V(2Y + 1) &= 2^2 V(Y) \\ &= 4(0.7270) = 2.9080 \end{aligned}$$